

MATH 121: COMMUTATIVE ALGEBRA

PRISHITA DHARAMPAL

1. RINGS

Important Example: Let X be a “geometric set”, and consider the ring

$$\text{Hom}(X, \mathbb{R}) = \{X \xrightarrow{f} \mathbb{R}\}$$

with pointwise addition and multiplication

$$(f + g)(x) = f(x) + g(x), \quad (fg)(x) = f(x)g(x)$$

$r \in \text{Hom}(X, \mathbb{R})$ is a unit if and only if there exists $g \in \text{Hom}(X, \mathbb{R})$ such that $fg = 1$ if and only if $\forall x \in X, (fg)(x) = f(x)g(x) = 1$

$$\text{Hom}(X, \mathbb{R})^\times = \text{Hom}(X, \mathbb{R}^\times)$$

Definition 1.1. Let $a \in R, a \neq 0$

- a is a unit if and only if $\exists b \in R$ such that $ab = 1$,
- a is a zero divisor if and only if $\exists b \in R \setminus \{0\}$ such that $ab = 0$,

Lemma 1.2. The set of units $R^\times = \{a \in R \mid a \text{ is a unit}\}$ is a group under multiplication.

Proof: Worksheet (03/31).

Lemma 1.3 (Useful Lemma). If $\phi : R \rightarrow S$ is a ring homomorphism, then it follows from the definition that

- $\phi(0_R) = 0_S$
- $\phi(-a) = -\phi(a)$
- $\phi(a^{-1}) = \phi(a)^{-1}$

Remark 1.4. $\mathbb{Z}$ is initial in the category of rings. There is a unique homomorphism from the initial object to every other object. For any ring R ,

$$\text{Hom}(\mathbb{Z}, R) = \{1 \rightarrow 1_R\}$$

We don't know anything about homomorphisms to the initial object.

2. IDEALS

Lemma 2.1. If $\phi : R \rightarrow S$ is any ring homomorphism and $J \subset S$ is an ideal $\phi^{-1}(J)$ is also an ideal.

Proof: Worksheet Q4.1.

Theorem 2.2. If $I \subset R$ is an ideal then,

- (1) There exists a ring R/I and a surjective ring homomorphism $\pi : R \rightarrow R/I$ such that $\ker(\pi) = I$. Moreover R/I and π are unique upto isomorphism.
(2) There is an inclusion preserving bijection

$$\{\text{Ideals in } R/I\} \leftrightarrow \{\text{Ideals in } R \text{ containing } I\}$$

$$\bar{J} \leftrightarrow \pi^{-1}(\bar{J})$$

Proof.

- (1) Construct R/I explicitly.
(2) I gives an equivalence relation $\sim$ on R such that

$$a \sim b \iff a - b \in I \iff a = b + I$$

Show that equivalence classes $R/\sim$ is a ring.

$$R \rightarrow R/I = R/\sim$$

$$a \mapsto \bar{a} = a + I$$

Let $\bar{J}$ be an ideal in R/I , and J be an ideal in R and define the maps

$$\phi : \bar{J} \rightarrow \pi^{-1}(\bar{J}), \quad \psi : J \rightarrow \pi(J)$$

Now we want to show that $\phi \circ \psi = \psi \circ \phi = id$. Consider $\phi \circ \psi(J)$ and $\psi \circ \phi(\bar{J})$. Note that

$$\psi \circ \phi(\bar{J}) = \pi(\pi^{-1}(\bar{J})) = \bar{J}$$

since π is surjective. Thus $\psi \circ \phi = id$.

Now, note that $J \subseteq \phi \circ \psi(J) = \pi^{-1}(\pi(J))$, and since $\bar{0} \in \pi(J)$, $I \subseteq \pi^{-1}(\pi(J))$. An element $x \in \pi^{-1}(\bar{J})$ looks like $x + I$, and $\pi^{-1}(\bar{J}) = J + I$. Moreover since $\bar{0} \in \bar{J}$, we have $\pi^{-1}(\bar{0}) \subseteq \pi^{-1}(\bar{J})$ which implies $I \subseteq J$. Thus, $\psi \circ \phi(J) = J$ and $\psi \circ \phi = id$.

□

Exercise: Describe the ideals in

- (1) $\mathbb{Z}/\langle 24 \rangle = \{\langle 0 \rangle, \langle 1 \rangle, \langle 2 \rangle, \langle 3 \rangle, \langle 4 \rangle, \langle 6 \rangle, \langle 8 \rangle, \langle 12 \rangle\}$
- (2) $\mathbb{Z}/\langle n \rangle = \{\langle d \rangle : d|n\}$
- (3) $\mathbb{Z}/\langle 3 \rangle = \{\langle 0 \rangle, \langle 1 \rangle\}$
- (4) $\mathbb{Z}/\langle 5 \rangle = \{\langle 0 \rangle, \langle 1 \rangle\}$
- (5) $\mathbb{C}[x, y]/\langle y - x^2 \rangle \cong \mathbb{C}[x] = \{\text{Ideals in } \mathbb{C}[x]\}$
- (6) $\mathbb{C}[x]/\langle x^n \rangle = \{\langle x^i \rangle : 0 \leq i < n\}$

Lemma 2.3. If $\phi : R \rightarrow S$ is a ring homomorphism with $\ker(\phi) \supseteq I$ then there exists a unique morphism $\tilde{\phi} : R/I \rightarrow S$

$$\begin{array}{ccc} R & \xrightarrow{\pi} & R/I \\ \phi \downarrow & \swarrow \tilde{\phi} & \\ S & & \end{array}$$

Universal Property of Quotients: Upto isomorphism R/I is the ring such that

$$\text{Hom}(R/I, S) = \{\phi : R \rightarrow S : I \subset \ker(\phi)\}$$

We want to define $\tilde{\phi}(\bar{a}) = \tilde{\phi}(a + I) = \phi(a)$ and $\tilde{\phi}(\bar{0}) = 0$.

Lemmact 2.4. Maximal Ideal $\implies$ Prime Ideals.

Definition 2.5. The spectrum of a ring R is the set

$$\text{Spec}(R) = \{\text{Prime Ideals in } R\}$$

the maximal spectrum of R is the set

$$m\text{Spec}(R) = \{\text{Maximal Ideals in } R\}$$

$\text{Spec}(R)$ is where the 'geometry lives'. Note that $m\text{Spec}(R) \subset \text{Spec}(R)$.

Example: $R = \mathbb{Z}[\sqrt{-5}]$. 2 is irreducible but not prime. $R/\langle 2 \rangle = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Example: $R = \mathbb{C}[x, y]/\langle x^3 - y^2 \rangle$. x is irreducible, but $\langle x \rangle$ is not prime. $R/\langle x \rangle = \mathbb{C}[y]/\langle y \rangle$ and y is a zero divisor.

Theorem 2.6. Let R be a non-zero ring.

- (1) R has a maximal ideal.
- (2) If $I \subsetneq R$ is a proper ideal there exists a prime ideal $P \subseteq R$ such that $I \subset P \subsetneq R$.

Proof. Zorn's Lemma: Let $(P, \leq)$ be a po-set.

If $P \neq \emptyset$ and if $C = C_1 \leq C_2 \leq C_3 \leq \dots$ is a chain in $(P, \leq)$ then there exists $B \in P$ such that $C_i \leq B$ for all i , then there exists $M \in P$ such that if $S \leq M$ then $M = S$.

Let $P = \{\text{Proper ideals in } R\}$ and $\leq$ be inclusion. We claim that $(P, \leq)$ is a poset.

- (1)
 - $P \neq \emptyset$ since $\langle 0 \rangle \in P$.
 - Given a chain in P

$$I_1 \subseteq I_2 \subseteq \dots$$

Then, define $B = \bigcup I_j$. Note that B is an ideals since union of nested ideals is an ideal. And B is proper since $1 \notin I_j$ for all j , and hence $1 \notin B$. Then there exists some ideal $M \subset R$ such that M is maximal.

- (2) Follows from (1). Let $R = R/I$. Pre-image of prime ideals is prime.

Thus, $m\text{Spec}(R) \neq \emptyset$ and $\text{Spec}(R) \neq \emptyset$. □

3. ZARISKI TOPOLOGY

Definition 3.1. If $S \subseteq R$ is any subset of R ,

$$\mathbb{V}(S) = \{[P] \in \text{Spec}(R) | S \subseteq P\} \subseteq \text{Spec}(R)$$

Definition 3.2. An algebraic subset of $\mathbb{K}^n$ of some field K is a set of the form

$$\mathbb{V}(S) = \{P \in \mathbb{K}^n | f(p) = 0, \forall f \in S\}$$

for a subset $S \subset \mathbb{K}[x_1, \dots, x_n]$

Examples:

- $\mathbb{V}(x^2 + y^2 - 1) \subseteq \mathbb{R}^2$

$$\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 - 1 = 0\}$$

- $\mathbb{V}(x^2 + y^2 - 1) \subseteq \mathbb{R}^3$

$$\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 - 1 = 0\}$$

- $\mathbb{V}(x^2 - y) \subseteq \mathbb{R}[x, y]$

$$\{(x, y) \in \mathbb{R}[x, y] \mid x^2 = y\}$$

- $\mathbb{V}(x^2 - xy) \subseteq \mathbb{R}[x, y]$

$$\{(x, y) \in \mathbb{R}[x, y] \mid x = 0, x - y = 0\}$$

- $\mathbb{V}(x^2 + 1) \subseteq \mathbb{R}[x]$

$$\mathbb{V}(x^2 + 1) = \emptyset$$

- $\mathbb{V}(x^2 + 1) \subseteq \mathbb{C}[x]$

$$\mathbb{V}(x^2 + 1) = \{i, -i\}$$

- $\mathbb{V}(x^2 + 4) \subseteq \mathbb{F}_3[x]$

$$\mathbb{V}(x^2 + 4) = \emptyset$$

- $\mathbb{V}(x^2 + 4) \subseteq \mathbb{F}_p[x]$

TODO:

- $\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) \subseteq \mathbb{R}[x, y]$

$$\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) = \mathbb{V}(x^3(x - y)^3) = \{(x, y) \in \mathbb{R}[x, y] \mid x = 0, x - y = 0\}$$

- $\mathbb{V}(n(x^2 + y^2 + z^2), n \in \mathbb{Z}) \subseteq \mathbb{R}^3$

TODO:

- $\mathbb{V}(\text{a collection of any linear polynomials in } \mathbb{K}[x_1, \dots, x_n])$

TODO:

Definition 3.3. Let $V \subseteq \mathbb{K}^n$ be an algebraic set. The defining ideal of V is

$$\mathbb{I}(V) = \{f \in \mathbb{K}[x_1, \dots, x_n] \mid f(p) = 0, \forall p \in V\}$$

Definition 3.4. If $\mathfrak{M} \subseteq R$ is a maximial ideal, the residue field of $\mathfrak{M}$ is $R / \mathfrak{M}$.

Definition 3.5. Let $\mathbb{K}$ be a field. A $\mathbb{K}$ algebra is the data of a ring R together with a fixed ring homomorphism

$$\alpha : \mathbb{K} \rightarrow R$$

Definition 3.6. If $(R, \mathbb{K} \xrightarrow{\alpha} R)$ and $(S, \mathbb{K} \xrightarrow{\beta} S)$ are $\mathbb{K}$ -algebras, a ring map $\phi : R \rightarrow S$ is a morphism of $\mathbb{K}$ -algebras iff

$$\begin{array}{ccc}
R & \xrightarrow{\phi} & S \\
\alpha \uparrow & \nearrow \beta & \\
\mathbb{K} & &
\end{array}$$

Lemmact 3.7. If R is k -algebra, R/I has a natural k -algebra structure compatible with the quotient map.

$$\begin{array}{ccc}
R & \xrightarrow{\pi} & R/I \\
\alpha \uparrow & \nearrow \exists! & \\
k & &
\end{array}$$

Definition 3.8. A valuation ring of a field $\mathbb{K}$ is a subring $R \subset \mathbb{K}$ with the property that for all $x \in \mathbb{K}^\times$ either $x \in R$ or $x^{-1} \in R$.

4. GRÖBNER BASIS

Remark 4.1. Ideal Membership Problem: Suppose $I \subseteq R[x_1, \dots, x_n]$ an ideal. How can we algorithmically determine if $f \in I$?

Definition 4.2. A ring R is Noetherian if and only if every ascending chain of ideals stabilizes (Ascending Chain Condition), i.e.

$$I_1 \subseteq I_2 \subseteq \dots$$

there exists $N \in \mathbb{Z}$ such that $I_k = I_{k+1}$ for all $k \geq N$.

Lemma 4.3. R is Noetherian if and only if every ideal is finitely generated.

Lemma 4.4 (Dickson's Lemma). Every monomial ideal in $\mathbb{K}[x_1, \dots, x_n]$ is finitely generated.

Theorem 4.5 (Hilbert's Basis Theorem). If R is a Noetherian ring, then $R[x]$ is also Noetherian.

5. NULLSTELLENSATZ

Theorem 5.1 (Hilbert's Nullstellensatz I). If $\mathbb{K}$ is algebraically closed, then every maximal ideal of $\mathbb{K}[x_1, \dots, x_n]$ is of the form $\langle x_1 - a_1, \dots, x_n - a_n \rangle$ for some point $(a_1, \dots, a_n) \in \mathbb{K}^n$.

Theorem 5.2 (Hilbert's Nullstellensatz II). If $\mathbb{K}$ is algebraically closed and $V = \mathbb{V}(\mathcal{S}) \subset \mathbb{K}^n$ is an algebraic set, for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, then $\mathbb{I}(V) = \sqrt{\langle \mathcal{S} \rangle}$.

Theorem 5.3 (Hilbert's Weak Nullstellensatz). If $\mathbb{K}$ is algebraically closed and $I \subset \mathbb{K}[x_1, \dots, x_n]$ is an ideal then $\mathbb{V}(I) = \emptyset$ if and only if $I = \langle 1 \rangle$.

Theorem 5.4 (Hilbert's Nullstellensatz III). Let $\mathbb{K}$ be an algebraically closed field. If $V \subset \mathbb{K}^n$ is an algebraic subset then the maps $\mathbb{I}_V$ and $\mathbb{V}_V$ induce inverse, inclusion-reversing bijections:

$$\begin{array}{ccc}
\left\{ \begin{array}{c} \text{algebraic subsets} \\ \text{in } V \end{array} \right\} & \begin{array}{c} \xrightarrow{\mathbb{I}_V} \\ \xleftarrow{\mathbb{V}_V} \end{array} & \left\{ \begin{array}{c} \text{radical ideals} \\ \text{in } \mathbb{K}[V] \end{array} \right\} \\
& 5 &
\end{array}$$

Remark 5.5. $\mathbb{I}$ and $\mathbb{V}$ properties:

- (1) If $\mathcal{S} \subset \mathcal{T}$ then $\mathbb{V}(\mathcal{T}) \subset \mathbb{V}(\mathcal{S})$
- (2)

$$\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle) = \mathbb{V}(\sqrt{\langle \mathcal{S} \rangle})$$

- (3) $\mathbb{I}(V)$ is radical.
- (4) $\langle \mathcal{S} \rangle \subset \sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(\mathbb{V}(\mathcal{S}))$
- (5) $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\mathbb{I}(\mathbb{V}(\mathcal{S})))$
- (6) If $fg \in \mathbb{I}(V)$ then $V \subseteq \mathbb{V}(g) \cap \mathbb{V}(f)$.

6. LOCALIZATION

Definition 6.1. A subset $U \subset R$ is **multiplicatively closed** if and only if

- (1) $1_R \in U$
- (2) and if $s, u \in U$ then $su \in U$.

Example 6.1.

- (1) $R = \mathbb{Z}, U = \{1, 3, 9, \dots\}$ is multiplicatively closed.
- (2) $R = \mathbb{Z}, U = \{1, 3, 9\}$ is not multiplicatively closed.
- (3) $R = \mathbb{R}[x, y], U = \{f \in R \mid f(1, 0) \neq 0\}$ is multiplicatively closed.
- (4) $R = \mathbb{R}[x, y], U = \{f \in R \mid f(1, 1) \neq 0, \text{ or } f(0, 0) \neq 0\}$ is not multiplicatively closed.

Definition 6.2. Let R be a ring and $U \subseteq R$ be a multiplicative closed set. The **localization** of R at U is a ring $U^{-1}R$ together with a ring map $\eta : R \rightarrow U^{-1}R$ that is universal in the following sense:

- (1) Given a ring map $\psi : R \rightarrow T$ such that $\psi(U) \subseteq T^\times$ there exists a unique ring map $\tilde{\psi} : U^{-1}R \rightarrow T$ such that

$$\begin{array}{ccc} R & \xrightarrow{\eta} & U^{-1}R \\ & \searrow \psi & \downarrow \tilde{\psi} \\ & & T \end{array}$$

Construction of $U^{-1}R$:

Consider $R \times U$ as a set (r_1, u_1) where $r_1 \in R, u_1 \in U$ “ $=$ ” $\frac{r_1}{u_1}$. Define an equivalence relation

$$(r_1, u_1) \sim (r_2, u_2) \iff t(u_2r_1 - u_1r_2) = 0, \quad t \in U$$

Note if R is a domain $t = 1_R$.

Define $\frac{r}{u}$ to be the equivalence class of (r, u) in $R \times U / \sim$.

Set $U^{-1}R = R \times U / \sim = \left\{ \frac{r}{u} \mid r \in R, u \in U \right\} / \sim$

Make $U^{-1}R$ a ring with

$$\left(\frac{r_1}{u_1}\right)\left(\frac{r_2}{u_2}\right) = \frac{r_1 r_2}{u_1 u_2}, \quad \frac{r_1}{u_1} + \frac{r_2}{u_2} = \frac{r_1 u_2 + r_2 u_1}{u_1 u_2}$$

Additive Identity: $0/1$, **Multiplicative Identity:** $1/1$

$$\eta : R \rightarrow U^{-1}R$$

Then

$$\begin{aligned} \ker(\eta) &= \{r \in R \mid \eta(r) = 0\} \\ &= \{r \in R \mid \frac{r}{1} = \frac{0}{1}\} \\ &= \{r \in R \mid t(r \cdot 1 - 1 \cdot 0) = 0, t \in U\} \\ &= \{r \in R \mid tr = 0, t \in U\} \end{aligned}$$

If R is a domain, $\ker(\eta) = \langle 0 \rangle \implies \eta$ is injective, unless $0 \in U$.

Example 6.2. TODO: Let R be any ring and $U = \{0, 1\}$. Then

$$U^{-1}R = \left\{\frac{r}{0} \mid r \in R\right\} \cup \left\{\frac{r}{1} \mid r \in R\right\}$$

Let $\frac{r}{1} \stackrel{?}{=} \frac{0}{1} \iff t(r - 0) = 0 \iff tr = 0$ for some $t \in U$. We can take $t = 0$ and make this always true (same argument $\frac{r}{0} = \frac{0}{1}$). Hence $U^{-1}R$ is the zero ring.

Lemma 6.3. Let $U \subseteq R$ be multiplicatively closed. If U contains a nilpotent then $U^{-1}R$ is the zero ring.

Definition 6.4. If $P \subseteq R$ is a prime ideal then $U = R \setminus P$ is multiplicatively closed. Here we write $U^{-1}R = R_P$ and say R **localized at** P .

Lemma 6.5.

$$PR_P = \left\{\frac{r}{u} \mid r \in P, u \in R \setminus P\right\} \subseteq R_P$$

is a maximal ideal.

Proof. Let $PR_P \subsetneq I \subseteq R_P$. Then for $r \in I \setminus PR_P$ with $r \notin P \implies \frac{r}{u}$ with $r, u \in R \setminus P \implies \frac{u}{r} \in R_P \implies \frac{r}{u}$ is a unit in $R_P \implies I = R_P$. **TODO:** □

Proposition 6.6. R_P is a local ring (i.e., PR_P is the unique maximal ideal in R_P).

Proof. If $I \subseteq R_P$ is a maximal ideal, then $I \cap R \setminus P \neq \emptyset$, that is I contains a unit, and hence $I = R_P$. □

Example 6.3. Let $f \in R$, and $U = \{1, f, f^2, \dots\}$, then

$$U^{-1}R = R\left[\frac{1}{f}\right] = \left\{\frac{r}{u} \mid r \in R, u \in U\right\} = \left\{\frac{r}{f^k} \mid r \in R, k \in \mathbb{Z}_{\geq 0}\right\}$$

that is $U^{-1}R$ is the ring of polynomials in $1/f$ with coefficients in R .

Lemma 6.7.

$$R \left[\frac{1}{f} \right] \cong \frac{R[x]}{\langle xf - 1 \rangle}$$

Proof. $R[x] \rightarrow R \left[\frac{1}{f} \right], x \mapsto \frac{1}{f}$ and $1 \mapsto \frac{1}{1}$. This is surjective because for every $rx^k \in U^{-1}R$ we have $rx^k \mapsto \frac{r}{f^k}$. To show injectivity we must show that $\ker(\varphi) \supseteq \langle xf - 1 \rangle$ is trivial

TODO: . □

Theorem 6.8. *There is an inclusion preserving bijection*

$$\begin{aligned} \left\{ \text{prime ideals of } U^{-1}R \right\} &\xleftrightarrow{1:1} \left\{ \text{prime ideals } P \subseteq R \text{ such that } P \cap U = \emptyset \right\} \\ P &\mapsto \eta(P) \end{aligned}$$

Let $Q \subseteq R$ be a prime ideal such that $Q \cap U \neq \emptyset$. Consider $\eta(Q) \cdot U^{-1}R = \langle \eta(Q) \rangle$. Let $r \in Q \cap U, \eta(r) \in \eta(Q) \subseteq \eta(Q) \cdot U^{-1}R$. Since $r \in U, \eta(r)$ is a unit in $U^{-1}R$, which implies

$$\eta(Q) \cdot U^{-1}R = U^{-1}R$$

i.e. it's not prime. **TODO:**

Theorem 6.9.

$$\left\{ \text{Spec}(U^{-1}R) \right\} \xleftrightarrow{1:1} \left\{ P \in \text{Spec}(R) \mid P \cap U = \emptyset \right\} \subseteq \text{Spec}(R)$$

$$P \mapsto \eta^{-1}(P)$$

Definition 6.10. Let R be a ring, then a basic open subset of $\text{Spec}(R)$ is of the form $D(f) = \{P \in \text{Spec}(R) \mid f \notin P\}$ for $f \in R$.

And,

$$D(f) = \text{Spec}(R) \setminus V(I) = \text{Spec}(R) \setminus V(\langle f \rangle)$$

The intersection of basic open sets $D(f) \cap D(g)$ is

$$D(f) \cap D(g) = \{P \in \text{Spec}(R) \mid f \notin P, g \notin P\} = \{P \in \text{Spec}(R) \mid fg \notin P\} = D(fg)$$

Remark 6.11. We know that there is a bijection,

$$\text{Spec} \left(R \left[\frac{1}{f} \right] \right) \leftrightarrow \{P \in \text{Spec}(R) \mid \{1, f, f^2, \dots\} \cap P \neq \emptyset\} = \{P \in \text{Spec}(R) \mid f \notin P\} = D(f)$$

Hence there is a bijection,

$$\text{Spec} \left(R \left[\frac{1}{f} \right] \right) \leftrightarrow D(f)$$

Theorem 6.12. Let $f \in R$ and $\eta : R \rightarrow R \left[\frac{1}{f} \right]$ be a localization map. The induced map

$$\text{Spec} \left(R \left[\frac{1}{f} \right] \right) \xrightarrow{\eta^\#} \text{Spec}(R)$$

is a homeomorphism onto $D(f)$.

Definition 6.13. *The nilradical of R is the set of nilpotents*

$$\mathfrak{N}(R) = \{f \in R \mid f^k = 0, \text{ for some } k\} = \sqrt{\langle 0 \rangle}$$

Proposition 6.14. $\text{Spec}(R) \cong \text{Spec}(R/\mathfrak{N}(R))$ induced by the quotient map

$$R \xrightarrow{\pi} R/\mathfrak{N}(R)$$

such that

$$\text{Spec}(R/\mathfrak{N}(R)) \xrightarrow{\pi^\#} \text{Spec}(R)$$

Proposition 6.15. $\mathfrak{N}(R) = \bigcap_{P \in \text{Spec}(R)} P$

Proof. ($\implies$)

Let $f \in \mathfrak{N}(R)$ so $f^n = 0$. Pick any prime ideal $P \subseteq R$, $f^n = 0 \in P \implies f \in P$ since P is prime.

($\impliedby$)

We show given $f \notin \mathfrak{N}(R)$ there exists a prime $P \subseteq R$ with $f \notin P$ (i.e. $D(f) \neq \emptyset$). □

7. NONZERODIVISORS

Remark 7.1. If M is a R -module we say an element $r \in R$ is a nonzerodivisor on $M \iff M \xrightarrow{\times r} M$ is injective $\iff r \notin \text{Ann}(m), \forall m \in M \iff (0 :_M r) = \{m \in M \mid rm = 0\} = 0$

Definition 7.2. Let M be a R -module. A sequence of elements $x_1, \dots, x_t \in R$ is a **regular sequence** on M if and only if

- (1) x_i is a nonzerodivisor on $M / \langle x_1, \dots, x_{i-1} \rangle M$
- (2) $M \neq \langle x_1 \dots x_t \rangle M$ (non-degeneracy condition)

Lemma 7.3. If $(R, \mathfrak{m})$ is a local ring, M is a finitely generated non-zero R -module, and $x_1, \dots, x_t \in \mathfrak{m}$ then the conditions (1) and (2) above are equivalent to just (1).

Proof. Nakayama □

Theorem 7.4 (Krull's Intersection Theorem). Let R be a Noetherian ring, $I \subseteq R$ an ideal, and M a finitely-generated R -module. There exists an element $r \in I$ such that

$$(1 - r) \bigcap_{k=1}^{\infty} I^k M = 0$$

Corollary 7.4.1. If $(R, \mathfrak{m})$ is a local Noetherian ring, $I \subsetneq R$ then

$$\bigcap_{k=1}^{\infty} I^k = 0$$

Theorem 7.5. Let $(R, \mathfrak{m})$ be a local ring, R is Noetherian, and M is a non-zero finitely generated R -module. If $x_1, \dots, x_t \in \mathfrak{m}$ then $x_1, \dots, x_t$ is a regular sequence on M if and only if every permutation is a regular sequence on M .

Proof. Let x, y be a regular sequence on M . We want to show that y, x is a regular sequence on M .

Assume $ym = 0$ for some $m \in M$. Then we have $y\bar{m} = 0 \in M/xM \implies \bar{m} = 0 \implies m = xm_1$ for some m_1 .

That is,

$$0 = ym = yxm_1 \implies ym_1 = 0 \in M \implies y\bar{m}_1 = 0 \in M/xM$$

Then by inducting on the m_i s, we get $m \in \bigcap_{k=1}^{\infty} \langle x \rangle^k M$. By KIT, this implies $m = 0$, a contradiction, and thus y is a NZD on M .

Now, assume $x\bar{m} = 0 \in M/yM$. Then, $xm = ym_1 \in M \implies y\bar{m}_1 = 0 \in M/xM \implies \bar{m}_1 = 0 \implies m_1 = xm_2 \in M$. So in M ,

$$xm = ym_1 = xym_2 \implies m = ym_2 \implies \bar{m} = 0 \in M/yM$$

Thus x is a nonzerodivisor on M/yM , and y, x is a regular sequence. □

Definition 7.6. Given an ideal $I \subseteq R$ the **height** of I is

$$\begin{aligned} ht(I) &= \sup \{d \in \mathbb{Z}_{\geq 0} \mid \text{there exists a chain of prime ideals } P_0 \subsetneq P_1 \subsetneq \cdots \subsetneq P_d \subseteq I\} \text{TODO: this breaks} \\ &= \dim R - \dim \mathbb{V}(I) \\ &= \dim R - \dim R/I \\ &= \sup \{ht(P) \mid P \subseteq I\} \text{TODO: this breaks} \\ &= \inf \{ht(P) \mid P \subseteq I\} \end{aligned}$$

Lemma 7.7. Let R be a ring

- (1) $\dim(R) = \sup \{ht(P) \mid P \in \text{Spec}(R)\}$
- (2) If $(R, \mathfrak{m})$ is local, then $\dim(R) = ht(\mathfrak{m})$
- (3) Let $I \subseteq R$, then $ht(I) = ht(\sqrt{I})$

Theorem 7.8. Let R be a finitely-generated k -algebra. If R is an integral domain then for any ideal $I \subseteq R$,

$$ht(I) = \dim(R) - \dim(R/I)$$

Definition 7.9. Given a Noetherian local ring $(R, \mathfrak{m}, k)$, its **embedding dimension** $\mu(R)$ is the minimal number of generators required for its maximal ideal $\mathfrak{m}$.

Theorem 7.10 (Krull's Principal Ideal Theorem). Let R be a Noetherian ring, $x \in R$. If $P \subseteq R$ is a prime ideal minimal over $\langle x \rangle$ then $ht(P) \leq 1$.

Corollary 7.10.1. Let R be a Noetherian ring, and ideal $I = \langle r_1, \dots, r_n \rangle \subseteq R$. If $P \subseteq R$ is a prime ideal minimal over I then $ht(P) \leq n$.

Theorem 7.11 (Krull's Height Theorem). If R is a Noetherian ring and $P \subseteq R$ is prime and minimal over $\langle f_1, \dots, f_r \rangle$ (among primes, wrt inclusion). Then $ht(P) \leq r$.

Corollary 7.11.1. If $(R, \mathfrak{m})$ is a Noetherian local ring with residue field $k = R/\mathfrak{m}$,

$$\dim(R) = ht(\mathfrak{m}) \leq \mu(\mathfrak{m}) = \dim_k(\mathfrak{m}/\mathfrak{m}^2)$$

Definition 7.12. A local ring $(R, \mathfrak{m})$ is **regular** if and only if $\mathfrak{m}$ can be generated by exactly $\dim(R)$ many elements $\iff \mu(\mathfrak{m}) = \dim(R)$

Definition 7.13. Let $(R, \mathfrak{m})$ be a local ring or $\dim(R) = d$. A **system of parameters (SOP)** for R is a sequence of elements $f_1, \dots, f_d \in \mathfrak{m}$ such that $\sqrt{\langle f_1, \dots, f_d \rangle} = \mathfrak{m}$.

Lemma 7.14. Let $(R, \mathfrak{m})$ be local. A sequence $f_1, \dots, f_d \in \mathfrak{m}$ is a SOP $\iff R/\langle f_1, \dots, f_d \rangle$ has dimension zero.

Lemma 7.15. Let x be a nonzerodivisor on a Noetherian ring R .

$$\dim R/x = \dim R - 1$$

Corollary 7.15.1. If $x_1, \dots, x_n$ is a regular sequence on R then

$$\dim(R/\langle x_1, \dots, x_n \rangle) = \dim R - n$$

Definition 7.16. Let R be a ring, $I \subseteq R$ an ideal, M a module, The depth of M with respect to I is

$$\text{depth}_I(M) = \{\text{maximal length of a regular sequence on } M \text{ contained in } I\}$$

If $(R, \mathfrak{m})$ is local,

$$\text{depth}(M) = \text{depth}_{\mathfrak{m}}(M)$$

Example 7.1. Depth is not always finite. For instance, if $R = M = \mathbb{K}[x_1, x_2, \dots]$ where $x_1, \dots, x_n$ is a regular sequence for all n .

Proposition 7.17. Let M be a R -module, and $x \in R$, then TFAE,

(1) x is a regular sequence on M .

(2) $k(x, M) = 0 \rightarrow M \xrightarrow{\times x} M \rightarrow 0$

Where $H_1(k(x, M)) = 0$ and $H_0(k(x, M)) \neq 0$. In general,

$$H_1(k(x, M)) = \ker(M \xrightarrow{\times x} M) = (0 :_M x)$$

and,

$$H_0(k(x, M)) = M/xM$$

If $x, y \in R$.

$$k(x, y, M) = 0 \rightarrow M \xrightarrow{\begin{bmatrix} y \\ -x \end{bmatrix}} M \oplus M \xrightarrow{\begin{bmatrix} x & y \end{bmatrix}} M \rightarrow 0$$

In general we need,

$$H_0(k(x, y, m)) = M/(xf + yg \mid f, g \in M) = M/\langle x, y \rangle M \neq 0$$

$$H_2(k(x, y, m)) = \ker \left(\begin{bmatrix} y \\ -x \end{bmatrix} \right) = \{m \in M \mid m(-x) = 0, my = 0\}$$

Theorem 7.18 (Noether). Let R be a Noetherian ring if $I \subsetneq R$ then the set $\{\mathfrak{p} \in \text{Spec}(R) \mid \mathfrak{p} \text{ is minimal over } I\}$ is finite.

Lemma 7.19 (Prime Avoidance). Let R be a ring and $I, J_1, \dots, J_n$ be proper ideals and let $J_2, \dots, J_n$ be prime ideals. Then,

(1) If $I \subseteq \bigcup_{i=1}^n J_i$ then $I \subseteq J_i$ for some i .

(2) If $I \not\subseteq J_i$ for all i then there exists $x \in I$ such that $x \notin \bigcup_{i=1}^n J_i$

Theorem 7.20. Let R be a Noetherian ring and M a finitely generated R -module then

$$\#\{\mathfrak{p} \in \text{Spec}(R) \mid \mathfrak{p} = \text{Ann}(m), m \in M\} < \infty$$

such primes are called associated primes of M .

Lemma 7.21. If $(R, \mathfrak{m})$ is a local Noetherian regular ring then R is an integral domain.

Proof. We will show this by induction on the dimension of R .

Base Case: $\dim(R) = 0$. Then,

$$\dim_R(\mathfrak{m} / \mathfrak{m}^2) = \dim(R) = 0$$

Then as vector spaces $\mathfrak{m} = \mathfrak{m}^2 = \mathfrak{m} \mathfrak{m}$. Then Nakayama's Lemma implies that $\mathfrak{m} = \langle 0 \rangle$, and hence R is a field and thus a domain.

Inductive Step: Assume $\dim(R) = d$, and the claim is true for all rings of $\dim \leq d - 1$. Let $(R, \mathfrak{m})$ be regular, i.e. $\dim(R) = \dim_R(\mathfrak{m} / \mathfrak{m}^2)$. We want to quotient R by an element $x \in R$ such that

- $R/\langle x \rangle$ has $\dim = d - 1$.

By KPT $R/\langle x \rangle$ has $\dim = d - 1 \iff x$ is a nonzerodivisor $\iff x$ is not contained in an associated prime of R . So we must avoid the finitely many associated primes.

- $R/\langle x \rangle$ is regular.

$R/\langle x \rangle$ is regular $\iff \dim_R(\mathfrak{n} / \mathfrak{n}^2) = \dim(R/\langle x \rangle) = d - 1$, where $\mathfrak{n} = \mathfrak{m} / \langle x \rangle$.

We claim if $x \notin \mathfrak{m}^2$ then $\dim(\mathfrak{n} / \mathfrak{n}^2) = d - 1$. By prime avoidance we may pick $x \in \mathfrak{m}$ such that $x \notin \mathfrak{m}^2$ and $x \notin$ an associated prime.

Thus $(R/\langle x \rangle)$ is a regular local Noetherian ring of dimension $d - 1$. So by induction it is a domain.

Now assume for contradiction that R is not a domain. Then

$$ab = 0 \quad \text{for } a, b \in \mathfrak{m}$$

reduction mod x gives us

$$\bar{a}\bar{b} = 0 \quad \text{in } R/\langle x \rangle$$

WLOG let $\bar{a} = 0 \implies a = a_1x$. Then since x is a nonzerodivisor

$$ab = (a_1x)b = 0 \implies a_1b = 0$$

We can repeat this process and get,

$$a \in \bigcap_{n \geq 1} \langle x^n \rangle$$

which implies $a = 0$ by KIT. Thus R is a domain. □

Theorem 7.22. *If $(R, \mathfrak{m})$ is a regular local Noetherian ring then*

$$\text{depth}_{\mathfrak{m}}(R) = \dim(R)$$

Proof. We will show this by induction on the dimension of R .

Base Case: $\dim(R) = 0$. Then R is a field and $\text{depth}_{\mathfrak{m}}(R) = \dim(R)$.

Inductive Step: Assume $\dim(R) = d$, and the claim is true for all rings of $\dim \leq d - 1$. Pick $x \in \mathfrak{m} \setminus \mathfrak{m}^2$. Since R is a domain, x is a nonzerodivisor. Then,

$$\dim(R/\langle x \rangle) = d - 1, \quad \dim(\mathfrak{n} / \mathfrak{n}^2) = d - 1$$

Now pick a regular sequence $\bar{y}_2, \dots, \bar{y}_d$ for $R/\langle x \rangle$. Pick lifts $y_2, \dots, y_d$ in R of $\bar{y}_2, \dots, \bar{y}_d$ look at $x, y_2, \dots, y_d$,

- x is a nonzerodivisor
- y_2 is a nonzerodivisor in since $y_2 = \bar{y}_2$ on $R/\langle x \rangle$.

We get the rest from $\bar{y}_2, \dots, \bar{y}_d$ being a regular sequence in $R/\langle x \rangle$. Thus $\dim(R) = \text{depth}_{\mathfrak{m}}(R)$. □

Definition 7.23. *A local ring $(R, \mathfrak{m})$ is **Cohen-Macaulay** if and only if*

$$\text{depth}(R) = \dim(R)$$

Corollary 7.23.1. *If $(R, \mathfrak{m})$ is a local Noetherian ring and R is regular then R is Cohen-Macaulay. The converse is not true.*

Definition 7.24. Let R be a ring. An R -module is projective if and only if given any surjective R -linear map $f : M \twoheadrightarrow N$ and any R -linear map $g : P \rightarrow N$

$$\begin{array}{ccc} & P & \\ \tilde{g} \swarrow & \downarrow g & \\ M & \xrightarrow{f} & N \end{array}$$

then there exists $\tilde{g}$ as above.

Example 7.2. (1) Let F be a free R -module. We claim that F is projective.

$$\begin{array}{ccc} & F & \\ \tilde{g} \swarrow & \downarrow g & \\ M & \xrightarrow{f} & N \end{array}$$

g is determined on a basis $\{e_i\}$. Then we define $\tilde{g}$ on lifts of $g(e_i)$.

- (2) Let $R = \mathbb{Z}/6\mathbb{Z}$ and $M = \langle 3 \rangle = \{\bar{0}, \bar{3}\}$. Then M is not free since free modules of R have either 1 element, 6^n elements, or infinite elements. We have $R \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$. And $M \cong \mathbb{Z}/2\mathbb{Z}$. So M is a summand of a free-module. Then by the theorem below, M is projective, not free.

Theorem 7.25. Let R be a ring and P an R -module. Then TFAE,

- (1) P is projective.
- (2) Every surjection $M \twoheadrightarrow P$ splits.
- (3) P is a direct summand of a free module.
- (4) the functor $\text{Hom}_R(P, -)$ is exact.

Proof. (1) $\implies$ (2)

Pick a surjection. Then,

$$\begin{array}{ccc} & P & \\ \swarrow & \downarrow \text{id} & \\ M & \xrightarrow{\quad} & P \end{array}$$

(2) $\implies$ (3)

Pick a surjection with F free

$$0 \rightarrow k \rightarrow F \twoheadrightarrow P \rightarrow 0$$

by (2) this splits on right and by splitting lemma $F \cong P \oplus k$.

(1) $\equiv$ (4)

Since $\text{Hom}_R(P, -)$ is injective.

(3) $\implies$ (1)

Assume $F \cong P \oplus k$ with F free.

$$\begin{array}{ccc} & & P \\ & & \downarrow g \\ M & \xrightarrow{f} & N \end{array}$$

Direct Sum-ing with k ,

$$\begin{array}{ccc} & & P \oplus k \cong F \\ & & \downarrow g \oplus \text{id} \\ M \oplus k & \xrightarrow{f \oplus \text{id}} & N \oplus k \end{array}$$

TODO: Then there exists a map $F \rightarrow M \oplus k$

Hence proved. □

Definition 7.26. A matrix $E \in \text{Mat}_{n \times n}(R)$ is **idempotent** if and only if $E^2 = E$.

Remark 7.27. If E is an idempotent then $I - E$ is an idempotent.

Lemma 7.28. Let R be a finitely generated R -module then P is projective if and only if there exists an idempotent matrix $E \in \text{Mat}_{n \times n}(R)$ for some $n > 0$ such that $P \cong \text{im}(R^n \xrightarrow{E} R^n)$.

Proof. ($\implies$)

Assume P is projective. Since P is finitely generated

$$0 \rightarrow k \rightarrow R^n \rightarrow P \rightarrow 0$$

Since $R^n \rightarrow P$ splits $R^n \cong P \oplus k$. Now let $E : R^n \rightarrow R^n$ be a projection onto P . (Projection is idempotent).

($\impliedby$)

Assume $P \cong \text{im}(R^n \xrightarrow{E} R^n)$ with $E^2 = E$. Then,

$$R^n \cong \text{im}(E) \oplus \ker(E)$$

and $\text{im}(E) \cong P$. □

Definition 7.29. A module M is **finitely-presented** if there exists an exact sequence

$$R^m \rightarrow R^n \twoheadrightarrow M \rightarrow 0$$

for some m, n .

Remark 7.30. If M is finitely-generated

$$\begin{array}{ccccccc} 0 & \longrightarrow & R^m & \longrightarrow & R^n & \twoheadrightarrow & M \longrightarrow 0 \\ & & \uparrow & & \nearrow & & \\ & & R^m & & & & \end{array}$$

Being finitely-presented is the same as being finitely-generated and the kernel of the surjective map being fg. **TODO:**

Theorem 7.31. Let R be a ring and let M be a finitely-presented R -module. The following are equivalent:

- (1) M is projective
- (2) $M_{\mathfrak{p}}$ is a free $R_{\mathfrak{p}}$ module for all $\mathfrak{p} \in \text{Spec}(R)$
- (3) $M_{\mathfrak{p}}$ is free as an $R_{\mathfrak{p}}$ module for all $\mathfrak{p} \in \text{mSpec}(R)$
- (4) $M[1/f_i]$ is free for $i = 1, \dots, n$ where $\langle f_1, \dots, f_t \rangle$

Theorem 7.32. Let $(R, \mathfrak{m})$ be a local ring. If P is a finitely-presented projective R -module then P is free.

Proof. Let $(R, \mathfrak{m}, k)$ be a local ring where k is the residue field and let P be a finitely-presented R module that is projective. Pick $y_1, \dots, y_t \in P$ whose images are a basis for $P/\mathfrak{m}P$ as a k -vector space (Nakayama). Then we have a surjection,

$$R^t \twoheadrightarrow P, \quad e_i \mapsto y_i$$

Since P is projective this map splits,

$$R^t \cong P \oplus k$$

for a finitely-generated module k . This gives us a map

$$R^n \twoheadrightarrow P \oplus k, \quad e_i \mapsto (y_i, 0)$$

Tensoring by $R/\mathfrak{m}$ gives us

$$k^n \twoheadrightarrow P/\mathfrak{m}P \oplus k/\mathfrak{m}k, \quad e_i \mapsto \overline{y_i}$$

Which implies $k/\mathfrak{m}k = 0$. By Nakayama's lemma applied to k ,

$$\implies k = 0 \implies R^t \cong P$$

□

Theorem 7.33 (Quillen-Suslin). If k is a field and M is a finitely-presented projective module over $k[x_1, \dots, x_n]$ then M is free.

Geometrically speaking this means "over $\mathbb{A}^n$ every algebraic" vector bundle of finite rank is trivial.

Definition 7.34. Let R be a ring and M be a R -module. A **projective resolution** of M is a chain complex

$$0 \leftarrow P_0 \xleftarrow{d_0} P_1 \xleftarrow{d_1} P_2 \xleftarrow{d_2} \dots$$

such that

- (1) The P_i are projective for all i .
- (2) $H_i(P_{\bullet}, d_{\bullet}) = \begin{cases} M & i = 0 \\ 0 & \text{otherwise} \end{cases}$

We say that this is a **free resolution** if P_i is free for all i .

Definition 7.35. The **projective dimension** of a module M is

$$\text{pd}(M) = \inf\{\text{lengths of projective resolutions of } M\}$$

Definition 7.36. A free resolution $F_\bullet$ of a module M is **minimal** if and only if for each differential

$$F_i \xleftarrow{d_i} F_{i+1}$$

we have that $\text{img}(d_i) \subseteq \mathfrak{m} F_i$

Definition 7.37. If $F_\bullet$ and $G_\bullet$ are chain complexes of R -modules, then an isomorphism $\psi_\bullet : F_\bullet \rightarrow G_\bullet$ is a collection $\{\psi_i : F_i \rightarrow G_i\}$ of R -module maps such that

(1) ψ_i is an isomorphism for all i and,

$$(2) \quad \begin{array}{ccc} F_i & \xleftarrow{d_i} & F_{i+1} \\ \downarrow \psi_i & & \downarrow \psi_{i+1} \\ G_i & \xleftarrow{\delta_i} & G_{i+1} \end{array}$$

Theorem 7.38. Let $(R, \mathfrak{m})$ be a Noetherian local ring and M a finitely-generated R -module, then

- (1) A minimal free resolution of M exists
- (2) If $F_\bullet$ and $G_\bullet$ are minimal free resolutions of M then $F_\bullet \cong G_\bullet$.

Proof. (1) Pick a minimal generating set of M $m_1, \dots, m_t$ such that

$$0 \leftarrow M \xleftarrow{\pi} F_\bullet \cong R^{\oplus t} \leftarrow \ker(\pi) \leftarrow 0$$

Since R is Noetherian and $F_\bullet$ is finitely-generated, $\ker(\pi)$ is finitely-generated. Pick a minimal generating set $k_1, \dots, k_s$ such that

$$\ker(\pi) \xleftarrow{\pi_1} F_1 \cong R^{\oplus s}, \quad k_i \leftarrow e_i$$

So we have,

$$\begin{array}{ccccccc} & & & & 0 & & \\ & & & & \uparrow & & \\ & & & & \ker(\pi) & & \\ 0 & \longleftarrow & M & \xleftarrow{\pi} & F_0 & \longleftarrow & \ker(\pi) \longleftarrow 0 \\ & & & \swarrow d_0 & \uparrow \pi_1 & & \\ & & & & F_i & & \\ & & & & \uparrow & & \\ & & & & \ker(\pi_2) & & \\ & & & & \uparrow & & \\ & & & & 0 & & \end{array}$$

such that $\text{coker}(d_0) \cong M$ and $\text{img}(d_0) \subseteq \mathfrak{m} F_0$ since none of $m_1, \dots, m_t$ lie in $\mathfrak{m} M$. The rest of the proof follows by induction. □

Theorem 7.39 (Kaplansky). Let $(R, \mathfrak{m})$ be a local ring and P is any R -module then P is projective if and only if P is free.

Definition 7.40. Let $F_\bullet$ and $G_\bullet$ be projective chain complexes a **chain homotopy** between chain maps $\psi_\bullet$ and $\phi_\bullet$ is the data of R -module maps $h_i : F_i \rightarrow G_{i+1}$ such that,

$$\phi_{i+1}(x) - \psi_{i+1}(x) = h_i(d_i(x)) + \delta_{i+1}(h_{i+1}(x))$$

$$\begin{array}{ccccc}
 F_i & \xleftarrow{d_i} & F_{i+1} & \xleftarrow{d_{i+1}} & F_{i+2} \\
 \psi_i \downarrow & & \downarrow \psi_{i+1} & & \downarrow \psi_{i+2} \\
 & \searrow h_i & & \searrow h_{i+1} & \\
 G_i & \xleftarrow{\delta_i} & G_{i+1} & \xleftarrow{\delta_{i+1}} & G_{i+2}
 \end{array}$$

Theorem 7.41. Let R be a ring, M and N R -modules and

$$0 \leftarrow M \xleftarrow{\pi_M} F_\bullet, \quad 0 \leftarrow N \xleftarrow{\pi_N} G_\bullet$$

be projective resolutions of M and N respectively. Given any R -module map $\phi : M \rightarrow N$ there exists a lift f_0 of ϕ to $F_\bullet$ and $G_\bullet$ i.e.

$$\begin{array}{ccccccc}
 0 & \leftarrow & M & \xleftarrow{\pi_M} & F_0 & \xleftarrow{d_0} & F_1 & \xleftarrow{d_1} & F_2 & \leftarrow & \dots \\
 & & \downarrow \phi & & \downarrow f_0 & & \downarrow f_1 & & \downarrow f_2 & & \\
 0 & \leftarrow & N & \xleftarrow{\pi_N} & G_0 & \xleftarrow{\delta_0} & G_1 & \xleftarrow{\delta_1} & G_2 & \leftarrow & \dots
 \end{array}$$

Moreover the lift f_0 is unique upto homotopy.

Corollary 7.41.1. Projective residue are unique upto chain homotopy.

Corollary 7.41.2. Say $(R, \mathfrak{m})$ is a local Noetherian ring and $F_\bullet$ and $G_\bullet$ are minimal free resolutions of a finitely-generated module M then

- (1) $F_\bullet$ and $G_\bullet$ are homotopy equivalent.
- (2) If $\phi_\bullet, \psi_\bullet : F_\bullet \rightarrow G_\bullet$ are chain maps that are homotopy equivalent then

$$\psi_\bullet \otimes R/\mathfrak{m} = \phi_\bullet \otimes R/\mathfrak{m}$$

Proof. Suppose h_i is a homotopy then

$$\phi_{i+1}(x) - \psi_{i+1}(x) = h_i(d_i(x)) + \delta_{i+1}(h_{i+1}(x))$$

the minimality of $F_\bullet$ implies $\text{img}(d_i) \subseteq \mathfrak{m} F_{i+1}$ and the minimality of $G_\bullet$ implies $\text{img}(\delta_i) \subseteq \mathfrak{m} G_{i+1}$. That is $\text{img}(\psi_{i+1} - \phi_{i+1}) \subseteq \mathfrak{m} G_{i+1}$. Tensoring by $R/\mathfrak{m}$ gives

$$F_{i+1}/\mathfrak{m} F_{i+1} \xrightarrow{\overline{\psi_{i+1} - \phi_{i+1}}} G_{i+1}/\mathfrak{m} G_{i+1}$$

is the zero map which implies that

$$\psi_{i+1} \otimes R/\mathfrak{m} = \phi_{i+1} \otimes R/\mathfrak{m}$$

□

Definition 7.42. The exterior product $\Lambda^{i+1}(R^n)$ is the free R -module with symbols $e_{j_1} \wedge e_{j_2} \wedge \cdots \wedge e_{j_{i+1}}$ modulo the alternating relations.

Definition 7.43. Let R be a ring and $x_1, \dots, x_n$ be a sequence of elements in R . The **Koszul complex** on $x_1, \dots, x_n$ is the chain complex of R -module $K_\bullet^R(x_1, \dots, x_n)$ where

$$K_{i+1}^R(x_1, \dots, x_n) = \Lambda^{i+1}(R^n) \xrightarrow{d_i} \Lambda^i(R^n) = K_i^R(x_1, \dots, x_n)$$

where

$$d_i(e_{j_1} \wedge e_{j_2} \wedge \cdots \wedge e_{j_{i+1}}) = \sum_{k=1}^{i+1} (-1)^k x_{j_k} \cdot e_{j_1} \wedge \cdots \wedge \widehat{e_{j_k}} \wedge \cdots \wedge e_{j_{i+1}}$$

Proposition 7.44. $K_\bullet^R(x_1, \dots, x_n)$ is a free chain complex.

Proof. We check $\delta_{i-1}(\delta_i(-)) = 0$ on basis element. And

$$\begin{aligned} \delta_{i-1}(\delta_i(e_{j_1} \wedge e_{j_2} \wedge \cdots \wedge e_{j_{i+1}})) &= \delta \left(\sum_{k=1}^{i+1} (-1)^k x_{j_k} \cdot e_{j_1} \wedge \cdots \wedge \widehat{e_{j_k}} \wedge \cdots \wedge e_{j_{i+1}} \right) \\ &= \sum_{k=1}^{i+1} (-1)^k x_{j_k} \cdot \delta(e_{j_1} \wedge \cdots \wedge \widehat{e_{j_k}} \wedge \cdots \wedge e_{j_{i+1}}) \\ &= \sum_{k=1}^{i+1} (-1)^k x_{j_k} \cdot \left(\sum_{\ell=1}^{i+1} (-1)^{\ell-1} x_{j_\ell} \cdot (e_{j_1} \wedge \cdots \wedge \widehat{e_{j_k}} \wedge \cdots \wedge \widehat{e_{j_\ell}} \wedge \cdots \wedge e_{j_{i+1}}) \right) \end{aligned}$$

TODO:

Then this is the zero map.

□

Lemma 7.45. $K_\bullet^R(x_1)$ is a free resolution $R/\langle x_1 \rangle$ if and only if x_1 is a nonzerodivisor if and only if x_1 is a regular sequence. Then the complex

$$0 \rightarrow R \xrightarrow{-x_1} R \rightarrow R/\langle x_1 \rangle \rightarrow 0$$

upon tensoring by M

$$0 \rightarrow M \xrightarrow{-x_1} M \rightarrow M/\langle x_1 \rangle M \rightarrow 0$$

remains exact.

Definition 7.46. Let R be a ring, M an R -module and $x_1, \dots, x_n$ a sequence of elements in R . The **koszul complex** of $x_1, \dots, x_n$ on M

$$K_\bullet^R(x_1, \dots, x_n) = K_\bullet^R(\bar{x}, M)$$

The **koszul homology** of $x_1, \dots, x_n$ on M is $H_i(\bar{x}, M) = H_i(K_\bullet^R(\bar{x}, M))$.

Theorem 7.47. Let R be a ring and M be a R -module. If $x_1, \dots, x_n$ is a regular sequence on M then

$$H_i(\bar{x}, M) = \begin{cases} 0 & i > 0 \\ M/\langle x_1, \dots, x_n \rangle M & i = 0 \end{cases}$$

Corollary 7.47.1. If $x_1, \dots, x_n$ is a regular sequence on R then the Koszul complex of $K_\bullet^R(x_1, \dots, x_n)$ is a free resolution of $R/\langle x_1, \dots, x_n \rangle$.

Theorem 7.48. Let $(R, \mathfrak{m})$ be a local ring and M a finitely-generated R -module. Let $x_1, \dots, x_n \in \mathfrak{m}$ then $x_1, \dots, x_n$ is a regular sequence on M if and only if $H_i(\bar{x}, M) = \begin{cases} 0 & i > 0 \\ M/\langle x_1, \dots, x_n \rangle M & i = 0 \end{cases}$

Definition 7.49. Let $(R, \mathfrak{m})$ be a local ring, $k = R/\mathfrak{m}$ be its residue field, and M be a R -module.

$$\mathrm{Tor}_R^i(k, M) = H_i(F_\bullet \otimes M)$$

where $F_\bullet$ is the free resolution of k as an R -module. And

$$\mathrm{Ext}_R^i(k, M) = H_i(\mathrm{Hom}_R(F_\bullet, M))$$

Remark 7.50 (Tor is Balanced). If $G_\bullet$ is a free resolution of M then

$$\mathrm{Tor}_R^i(M, k) = H_i(G_\bullet \otimes k) \cong H_i(F_\bullet \otimes M) \cong \mathrm{Tor}_R^i(k, M)$$

Lemma 7.51. Let $H_\bullet$ be a free resolution of M and $G_\bullet$ be a minimal free resolution of M then

$$H_\bullet \cong G_\bullet \oplus C_\bullet$$

That is, $\mathrm{pd}(M) = \text{length of the minimal free resolution}$.

Lemma 7.52. Let M be a finitely-generated module over a Noetherian local ring $(R, \mathfrak{m})$ with residue field k . Then

$$\mathrm{pd}_R(M) = \sup\{i \in \mathbb{Z}_{\geq 0} \mid \mathrm{Tor}_R^i(M, k) \neq 0\}$$

Proof. $\sup\{i \in \mathbb{Z}_{\geq 0} \mid \mathrm{Tor}_R^i(M, k) \neq 0\} \leq \mathrm{pd}_R(M)$. Compute $\mathrm{Tor}(k, M)$ by the minimal free resolution of M .

$$\mathrm{Tor}_R^i(M, k) = H_i(G_\bullet \otimes R) = H_i(G_\bullet \otimes R/\mathfrak{m}) = \dim_R(G_i/\mathfrak{m}G_i)$$

TODO:

□

Theorem 7.53 (Auslander-Buchsbaum Formula). Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring, and let M be a finitely-generated R -module. If $\mathrm{pd}_R M < \infty$ then,

$$\mathrm{pd}_R(M) + \mathrm{depth}_{\mathfrak{m}}(M) = \mathrm{depth}_{\mathfrak{m}}(R)$$

Definition 7.54. The standard $\mathbb{Z}$ -grading on $S = k[x_0, \dots, x_n]$ is a decomposition of S as an abelian group

$$S \cong \bigoplus_{d \in \mathbb{Z}} S_d$$

where S_d is the k -vector space spanned by monomials of degree k .

A graded module over S is an S -module with a decomposition as an abelian group

$$M \cong \bigoplus_{d \in \mathbb{Z}} M_d$$

such that $S_e \cdot M_d \subseteq M_{e+d}$.

Theorem 7.55 (Hilbert's Syzygy Theorem). *If M is a finitely-generated graded module over S then M has a free resolution of length at most $(n + 1)$.*

Theorem 7.56 (Eisenbud). *Let $(S, \mathfrak{m})$ be a regular local Noetherian ring and $f \in \mathfrak{m}^2$, $f \neq 0$ and is not a zero-divisor. If M is a finitely-generated module over $R = S/\langle f \rangle$ then the minimal R -free resolution of M is eventually 2-periodic*

$$0 \leftarrow M \leftarrow F_0 \leftarrow \cdots \leftarrow F_t \leftarrow R^s \xleftarrow{A} R^s \xleftarrow{B} R^s \xleftarrow{A} R^s \xleftarrow{B} \cdots$$

for some matrices A and B such that

$$AB = BA = f \text{ Id}$$

Theorem 7.57 (? i think this is a definition lol). *Let $(R, \mathfrak{m}, k)$ be a local Noetherian ring and M a finitely-generated R -module. The poincare series of M*

$$P_M(t) = \sum_{i \geq 0} \dim_R \text{Tor}_i^R(M, R) t^i$$

where $\dim \text{Tor}_i^R(k, k) = \binom{n}{i}$ and

$$P_R(t) = \sum_{i \geq 0} \binom{n}{i} t^i = (1 + t)^n$$

Theorem 7.58 (Auslander-Buchsbaum-Serie). *Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $k = R/\mathfrak{m}$. Then TFAE,*

- (1) R is a regular ring.
- (2) The global dimension of R is finite.
- (3) Every finitely-generated R -module has a projective resolution of finite length.
- (4) k has a projective resolution of length equal to $\dim(R)$.

Theorem 7.59 (Auslander-Buchsbaum-Serie 2). *Let $(R, \mathfrak{m})$ be a Noetherian local ring with residue field $k = R/\mathfrak{m}$. Then TFAE,*

- (1) R is a regular ring.
- (2) The global dimension of R is finite.
- (3) Every finitely-generated R -module has a projective resolution of finite length.
- (4) $P_R(t)$ is a polynomial.